

TECHNICAL WRITING PORTFOLIO · API DOCUMENTATION

Open-Meteo Weather API

Quick-Start Guide

Requests · Parameters · JSON Responses · Troubleshooting

For Developers and Technical Users

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Technical Writing Portfolio

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*Independent portfolio project based on publicly available Open-Meteo documentation.
This document is not official Open-Meteo documentation.*

1. Introduction

The Open-Meteo Weather Forecast API gives programmatic access to weather forecast data for specified geographic locations. Users can submit geographic coordinates and weather variables to receive forecast data in JSON format. This quick-start guide demonstrates how to create a basic forecast request, set request parameters, send the request, and interpret the returned JSON data.

Intended Audience

This guide is designed for developers and technical users who are new to the Open-Meteo Weather Forecast API. A working knowledge of HTTP requests and JSON is helpful but not required.

2. Base URL and Endpoint

The Open-Meteo Weather Forecast API uses HTTP GET requests to obtain forecast data.

Base URL

```
https://api.open-meteo.com
```

Forecast Endpoint

```
GET /v1/forecast
```

Combine the base URL and Endpoint to construct the request URL

```
https://api.open-meteo.com/v1/forecast
```

Geographic coordinates are required in a forecast request. Additional parameters can be added to specify the weather data returned in the response.

3. Request Parameters

Request parameters indicate the location and weather information included in a forecast request. The `latitude` and `longitude` parameters are mandatory. Optional parameters can be used to specify weather variables, units, time zones, and forecast length.

Parameter	Type	Required	Description
latitude	Float	Yes	Indicates the latitude of the forecast location.
longitude	Float	Yes	Indicates the longitude of the forecast location.
current	String array	No	Indicates the weather variables returned for current conditions.
hourly	String array	No	Indicates the weather variables returned as hourly forecast data.
temperature_unit	String	No	Indicates the temperature unit used in the response.
timezone	String	No	Indicates the time zone used for returned timestamps.
forecast_days	Integer	No	Indicates the number of forecast days returned.

4. Send Your First Request

Construct a forecast request by adding parameters to the request URL. Begin the query string with a question mark (?) and separate each additional parameter with an ampersand (&).

The following request retrieves the current temperature for the specified location:

```
GET https://api.open-meteo.com/v1/forecast?latitude=40.71&longitude=-74.01&current=temperature_2m
```

This request includes the following parameters:

- `latitude=40.71` specifies the latitude.
- `longitude=-74.01` specifies the longitude.
- `current=temperature_2m` requests the current air temperature at 2 meters above ground.

Send the request to the API using a web browser, API client, or application. A successful request returns the requested weather data in JSON format.

5. Understand the JSON Response

A successful request returns forecast data as a JSON object. The response includes information about the specified location and the current weather variable selected in the request.

The following example displays a simplified response:

```
{
  "latitude": 40.71,
  "longitude": -74.01,
  "current_units": {
    "temperature_2m": "°C"
  },
  "current": {
    "temperature_2m": 18.5
  }
}
```

In this response:

- `latitude` and `longitude` specify the geographic coordinates of the forecast location.
- `current_units` identifies the unit used for the returned current weather variable.
- `current` contains the requested current weather data.
- `temperature_2m` contains the air temperature at 2 meters above ground.

6. Errors and Troubleshooting

If a request contains an invalid parameter or value, the API returns an error response that describes the issue.

An error response may contain the following fields:

```
{
  "error": true,
  "reason": "Description of the error"
}
```

Use the `reason` field to identify the cause of the error and review the request for incorrect or unsupported parameters.

Common issues include:

Issue	Recommended Action
Missing coordinates	Verify that the request includes both <code>latitude</code> and <code>longitude</code> .
Invalid parameter	Check the parameter name against the API documentation.
Invalid parameter value	Verify that the value uses the format or range supported by the parameter.
Missing weather variable	Specify a supported variable using <code>current</code> , <code>hourly</code> , or another applicable parameter.

7. Additional Resources

For complete parameter descriptions, supported weather variables, and additional API options, refer to the official Open-Meteo documentation.

Open-Meteo Weather Forecast API Documentation

<https://open-meteo.com/en/docs>

Open-Meteo Website

<https://open-meteo.com>

PORTFOLIO NOTE

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